Prepare buffers (Volumes above are indicated per sample)

10x stocks of Wash buffer base (no spermidine/Tween) and Labeling buffer base (no spermidine/SAM/Hia5)

2 mL Wash buffer -> take **488 u**L and add Digitonin + 50XProtInh to make **Lysis buffer**
50 uL **labeling buffer** / 250.000 cells (typically ~ 1E6 cells -> **200 uL**)

Make 25 mM **spermidine stocks fresh monthly** and store at -20deg.
Add 2uL spermidine (liquid, ~6.38M) to 350 uL 0.1M HCl and 160 uL H2O – **check pH with strips!**

Protocol 6mA labeling and HMW DNA extraction

1) Place the Dounce homogenizer and pestles on ice, chill the centrifuge to 4 °C and preheat the ThermoMixer to 37°C..
2) **Carefully** transfer 3-4 (2mm diameter) tissue punch biopsies (~25mg) to the chilled Dounce homogenizer. Keep the Dounce homogenizer on ice during the entire disruption process
3) Add 500 uL of the Fiber-Seq Lysis buffer and let the tissue thaw for 1 minute.
4) Gently homogenize the tissue 10X with pestle A and 10X with B.

- Push the tissue with the pestle firmly into the bottom of the Dounce chamber with each stroke (Down + Up = 1X).
- Keep the tissue between tip of pestle and the bottom of the Dounce chamber for thorough homogenization.
- Homogenate may become foamy, but this is not a cause for concern. In the next step, transfer any foam that forms.

5) Incubate on ice for 5min before adding 1000uL of Wash buffer.
6) Transfer the lysate to a 2mL Protein LoBind microcentrifuge tube.
7) Rinse the pestles and homogenizer with the remaining 500 uL Wash buffer and add to the sample.
8) Pellet homogenate by centrifuging at 700 x g and 4 °C for 5 min. Discard supernatant.
9) Resuspend the pellet in 200uL of Hia5 labeling buffer – use a 1mL or wide bore tip.
10) Incubate on a ThermoMixer at 37deg and 900 rpm for 30 min.
11a) **Continue with Circulomics CBB Tissue protocol from step 8 onwards**
11b) **Continue with NEB Monarch HMW**

Protocol Nanopore sequencing (LSK-110, PromethION)

According to ONT protocol *Genomic DNA by Ligation (SQK-LSK110)* with the following modifications:

1) Start with 3-4ug of HMW DNA in 150 uL and sheer 25x with a 26G needle or in Megaruptor to 35kb.
2) Adjust volumes end-prep and FFPE repair step accordingly (*i.e.* vol x 3), omit control strand (CS)
3) Extend end-prep and FFPE repair steps from 5 to 30min (i.e. 30min at 20deg and 30min at 65deg)
4) Extend ligation step to 1h at RT
5) Elute the AMPure cleanups for 10 minutes and 20 minutes after the end-prep and ligation steps.
6) This should yield a ~3x library, aim to load near the high end of the 5-50fmol range, typically ~8uL

Note 1: library prep yield is typically 30-50%
Note 2: the amount loaded can be reduced during subsequent flushes to balance seq yield with # flushes

Fiber-seq brain tissue (BA24/MB) ASAP

10X Wash/Lysis base (10X-WLB) (200uL / sample)

	Stock		Final for 10X		V (uL)
Tris-HCl pH 7.4	1000	mM	100	mM	100
NaCl	5000	mM	100	mM	20
Water					880
TotVol					1000

10X Hia5 labeling base (10X-H5B) (20uL / sample)

	Stock		Final for 10X		V (uL)
Tris-HCl pH 8	1000	mM	150	mM	150
NaCl	5000	mM	150	mM	30
KCl	1000	mM	600	mM	600
EDTA pH 8	500	mM	10	mM	20
EGTA pH 8	250	mM	5	mM	20
Water					180
TotVol					1000

1X Wash buffer

	Stock		Final		V (uL)	X5
10X WLB	10	X	1	X	230	1150
BSA	10	%	0.1	%	23	115
Spermidine pH 7.4	25	mM	0.5	mM	46	230
Tween-20	10	%	0.1	%	23	115
Water					1978	9890
TotVol					2300	11500

1X Lysis buffer

	Stock		Final		V (uL)	X5
Wash buffer					488	2440
Digitonin	5	%	0.02	%	2	10
ProteaseInh	50	X	1	X	10	50
TotVol					500	2500

1X Hia5 labeling buffer

	Stock		Final		V (uL)	X5
10X-H5B	10	X	1	X	20	100
BSA	10		0.1		2	10
Spermidine pH 7.4	25		0.5		4	20
SAM	32		0.8		5	25
Hia5 enzyme	250		5		4	20
Water					165	825
					200	1000